Chapter 4. Decision Trees and Random Forests

Predictive Modeling, Lecture 3 (CART, impurity, pruning, bagging, OOB, random forest, importance)

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1 After the Linear Baseline: Where Does the Relation Change?

The language of linear models expresses effects as a global linear combination; to add interactions or nonlinearity a person must design the terms. The language of trees splits the input space by if–then conditions and uses a simple constant prediction in each region. Typical questions that trees answer naturally: in which customer group does the price effect change? Does risk jump beyond a threshold? Do different combinations of variables need entirely different rules? **A tree is a flexible baseline that discovers interactions and thresholds; a random forest is the ensemble that stabilizes this flexibility.**

2 The Tree as a Piecewise-Constant Function

Motivation. A regression tree splits the input space into rectangles and predicts a constant on each. This view explains both its strengths (interactions and nonlinearity for free) and its weakness (high variance from greedy, discontinuous splits).

Definition: regression tree

$$\hat{f}(x) = \sum_{m=1}^{M} c_m \, \mathbb{I}(x \in R_m),$$

where $R_1, \ldots, R_M$ are the leaf regions (a partition of $\mathcal{X}$), c_m is the prediction in region m, and the boundaries are axis-aligned splits by a threshold on one variable.

Within a leaf the model is very simple, but with enough leaves the whole model can express nonlinearity and interactions. The price: if one split changes, every rule below it can change, so a single tree has high variance. The same tree has two views: in feature space each rectangle is a leaf; in the condition structure each root-to-leaf path is an if–then rule.

2.1 Nested conditions create interactions automatically

Two splits already give

$$\hat{f}(x) = \begin{cases} c_1, & x_1 < s_1, \\ c_2, & x_1 \geq s_1 \ \wedge \ x_2 < s_2, \\ c_3, & x_1 \geq s_1 \ \wedge \ x_2 \geq s_2. \end{cases}$$

Here the effect of x_2 exists only when $x_1 \geq s_1$: an interaction. A linear model needs a term such as $x_1 x_2$ inserted by hand, whereas the tree's split path produces it. Example: a high-price product's discount effect differs between new and existing customers; one rule path captures both conditions.

2.2 Leaf predictions

Theorem: optimal leaf constants

For a fixed partition, (i) under squared loss the optimal c_m is the leaf mean $\hat{c}_m = \frac{1}{N_m} \sum_{i: x_i \in R_m} y_i$;
(ii) under absolute loss it is the leaf median;
(iii) for classification, the leaf class proportions $\hat{p}_{mk} = \frac{1}{N_m} \sum_{i: x_i \in R_m} \mathbb{I}(y_i = k)$ minimize the log loss, and $\hat{y}(x) = \arg\max_k \hat{p}_{mk}$ minimizes misclassification.

Proof [Supplement]. The loss separates over leaves, and within a leaf the problem is the constant-prediction problem of Ch. 1 applied to the empirical distribution of the leaf (mean, median, class frequencies). ■

Intuitively, the tree divides customers into small groups by conditions and gives everyone in a leaf the same demand, risk or price prediction. At leaf boundaries the prediction jumps like a staircase; if a smooth relation is required, keep other models as candidates. The majority class is the default label but $\hat{p}_{mk}$ can be used as a probability estimate — before converting it into actions check calibration and thresholds separately; proportions in small leaves are very unstable.

3 Growing the Tree: CART

3.1 One variable, one threshold per node

$$R_1(j,s) = \{x : x_j < s\}, \qquad R_2(j,s) = \{x : x_j \geq s\}.$$

All candidate variables j and cut points s are examined; the best binary split of the current node is chosen; the procedure repeats in the child nodes. The tree does not search for the global optimum at once; it is **greedy**, choosing the best split at the current step. One axis-aligned split is simple, but stacking several builds rectangular regions and complex interactions.

3.2 Regression split criterion

$$\Delta_{\text{RSS}}(j,s) = \sum_{i \in R} (y_i - \bar{y}_R)^2 - \sum_{r \in \{1,2\}} \sum_{i \in R_r(j,s)} (y_i - \bar{y}_{R_r})^2.$$

Choose (j,s) with the largest Δ_{RSS}, i.e. the split that makes the targets inside the children most homogeneous.

Derivation: efficient search and the ANOVA form of the gain

For node R with $N = N_1 + N_2$, the within/between decomposition gives

$$\Delta_{\text{RSS}} = N_1(\bar{y}_{R_1} - \bar{y}_R)^2 + N_2(\bar{y}_{R_2} - \bar{y}_R)^2 = \frac{N_1 N_2}{N}(\bar{y}_{R_1} - \bar{y}_{R_2})^2 \ \geq 0.$$

For a continuous x_j, cut points between consecutive sorted values are the only candidates; after sorting, running sums $\sum y$ and counts update in $O(1)$ per candidate, so all cut points of a variable cost $O(N \log N)$. A categorical variable with K levels has $2^{K-1} - 1$ binary partitions; for regression (and binary classification) it suffices to sort the levels by their mean target and treat them as ordered, leaving $K - 1$ candidates.

A good split now does not guarantee global optimality once later splits are considered.

3.3 Classification split criterion

$$G(R) = 1 - \sum_{k=1}^{K} p_{Rk}^2 \ \ \text{(Gini)}, \qquad H(R) = -\sum_{k=1}^{K} p_{Rk} \log p_{Rk} \ \ \text{(entropy)}.$$

A leaf with one class has impurity 0. A good split strongly reduces the weighted average impurity $\frac{N_1}{N} I(R_1) + \frac{N_2}{N} I(R_2)$. Gini and entropy usually give similar trees in practice.

Theorem: interpretation and monotonicity of impurity

(i) $G(R)$ is the probability that two labels drawn independently from the leaf distribution differ, equivalently the misclassification rate of a rule that labels at random according to p_R. (ii) Gini and entropy are strictly concave functions of p; hence any split satisfies $\frac{N_1}{N} I(R_1) + \frac{N_2}{N} I(R_2) \leq I(R)$, with equality iff both children have the parent's class proportions.

Proof [Supplement]. (i) $P(\text{two draws differ}) = 1 - \sum_k p_k^2$.

(ii) The parent's proportions are the weighted average $p_R = \frac{N_1}{N} p_{R_1} + \frac{N_2}{N} p_{R_2}$; apply Jensen's inequality to the concave impurity. ■

The misclassification rate $1 - \max_k p_k$ is concave but piecewise linear, so many splits give zero improvement; this is why smoother Gini/entropy are used for growing. Distinguish the split criterion (an internal training objective) from the final evaluation by the problem's metric (AUC, log loss, cost).

3.4 Many candidate splits cause selection bias

Continuous variables offer many cut points; categorical variables offer many partitions, and the more levels a variable has the more chances to split. Statistically, the split that reduced training loss most need not be the best for generalization, especially when many candidates are searched in small nodes. Remedies: a minimum leaf size, pruning, external validation; check high-cardinality identifier variables for leakage first.

3.5 Greedy CART: fast but not globally optimal

Greedy growth chooses the split with the largest immediate gain and never revisits earlier choices. Strengths: handles continuous and categorical variables, nonlinearity and interactions naturally. Cost: one early split can lock in the later structure. Do not treat a tree's structure as "the discovered truth"; look at validation performance and stability together.

3.6 Axis-aligned splits approximate oblique boundaries by staircases

Well suited: threshold rules such as "fewer than 5 uses in the last 30 days and at least one payment failure". Hard: diagonal boundaries like $x_1 + x_2 > c$ or smooth functions need many rectangles. Growing deeper to gain expressiveness makes leaves smaller and variance larger, so a tree is not always best for nonlinear relations. Compare linear baselines, linear models with transformations and tree ensembles under the same validation protocol.

4 Controlling Complexity

A deeper tree always lowers training error but test error can rise again: depth increases expressiveness but also the risk that small leaves and complex boundaries memorize accidental patterns.

Approach	Example	Pros and cons
pre-pruning	maximum depth, minimum leaf size, minimum split gain	fast, but stopping too early can miss good structure
post-pruning	grow large, then remove branches	sees a wider set of candidates, but requires a separate complexity choice
validation-based choice	choose depth / leaf size / pruning strength by CV	targets generalization; must be done inside the pipeline

Not "never grow deep trees", but *choose depth by generalization performance.*

4.1 Minimum leaf size is a statistical regularizer

$$\hat{p}_m = \frac{n_{m,+}}{N_m}, \qquad \mathrm{SE}(\hat{p}_m) \approx \sqrt{\frac{\hat{p}_m(1-\hat{p}_m)}{N_m}}.$$

Too small a leaf: local means and class proportions are driven by a few observations, so training looks good while new cases wobble (e.g. $N_m = 5$, $\hat{p} = 0.4$ gives SE ≈ 0.22). Too large a leaf: different groups are mixed, local thresholds and interactions are missed, bias increases. The minimum leaf size is not merely a speed option; it controls the stability of local means and probabilities.

4.2 Cost-complexity pruning

$$R_\alpha(T) = R(T) + \alpha|T|,$$

where $R(T)$ is the training loss or impurity-based risk of tree T, $|T|$ the number of leaves and α the price of keeping one more leaf. A split that lowers training loss only a little is pruned once α exceeds its benefit. α is chosen by CV like any other hyperparameter.

Derivation: weakest-link pruning

For an internal node t with subtree T_t, collapsing T_t to a leaf changes R_α by $\{R(t) - R(T_t)\} - \alpha(|T_t| - 1)$. Collapsing pays off as soon as

$$\alpha \geq g(t) = \frac{R(t) - R(T_t)}{|T_t| - 1}.$$

Repeatedly collapsing the node with the smallest $g(t)$ yields a nested sequence $T_0 \supset T_1 \supset \cdots \supset \{\text{root}\}$ with breakpoints $\alpha_1 < \alpha_2 < \cdots$; for every α the optimal subtree is in this sequence (Breiman et al.). CV then only needs to choose among finitely many trees.

5 From One Tree to Many: Bagging

5.1 A single tree has high variance

Resampling experiment: changing a few training observations can change the root split, leaf boundaries, selected variables and predictions; observations near a boundary can flip an early split, and deep trees are more sensitive to small-sample accidents. With MSE = Bias^2 + Var + σ^2, the goal of an ensemble is to average several slightly different trees and cancel the accidental wobble of each. **Trees are ensembled not because they are weak, but because they are strong yet unstable.**

5.2 Bagging

Algorithm: bagging (bootstrap aggregating)

For $b = 1, \ldots, B$: draw a bootstrap sample of size n with replacement; grow a deep tree $\hat{f}_b$. Aggregate by averaging (regression) or by averaging probabilities / majority vote (classification).

For a single tree one limits depth or prunes to lower variance, since that one tree is the final predictor. For bagging and random forests, individual trees are grown fairly deep for low bias and averaging reduces variance; overly strong pre-stopping would also cut the expressiveness of the ensemble. Not unlimited, though: with small samples, rare classes, probability quality or compute cost, minimum leaf size, maximum depth and weights still matter. Readability of an individual tree and predictive stability of the ensemble are different goals.

5.3 A bootstrap sample contains about 63.2% of the distinct observations

Theorem: inclusion probability

The probability that a given observation never appears in a bootstrap sample of size n is $(1 - 1/n)^n \to e^{-1} \approx 0.368$. Hence on average about $1 - 0.368 = 0.632$ of the distinct observations are used to grow one tree.

Proof [Supplement]. Each of the n independent draws misses the observation with probability $1 - 1/n$. Moreover $\log(1 - 1/n)^n = n \log(1 - 1/n) = -1 - \frac{1}{2n} + O(n^{-2}) \to -1$. ■

Some observations are drawn several times, others are left out; the left-out (out-of-bag, OOB) observations allow an internal evaluation.

5.4 Averages are sensitive to correlation between trees

Theorem: variance of an average of correlated predictors

If each $\hat{f}_b(x)$ has variance σ^2 and every pair has correlation $\rho \geq 0$, then

$$\operatorname{Var}\Big(\frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)\Big) = \sigma^2\Big[\rho + \frac{1-\rho}{B}\Big].$$

Proof [Supplement]. $\operatorname{Var}(\sum_b \hat{f}_b) = B\sigma^2 + B(B-1)\rho\sigma^2$. Divide by B^2: $\frac{\sigma^2}{B} + \frac{B-1}{B}\rho\sigma^2 = \sigma^2\{\rho + (1-\rho)/B\}$. ■

Increasing B shrinks the second term, but if ρ is large the average still wobbles: the variance floor is $\rho\sigma^2$. Averaging does not change the bias: $\mathbb{E}[\frac{1}{B}\sum_b \hat{f}_b] = \mathbb{E}\hat{f}_1$. Random forests add feature randomness precisely to lower ρ.

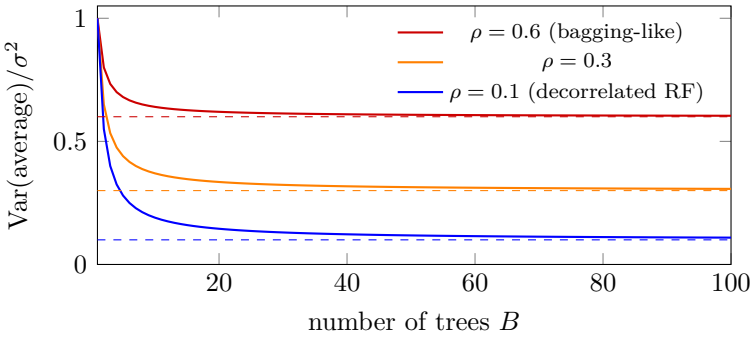


Figure 1: $\rho + (1-\rho)/B$: adding trees quickly removes the second term, but the variance of the ensemble levels off at the floor $\rho\sigma^2$ (dashed). Lowering the correlation between trees lowers the floor itself.

Task	Output of one tree	Ensemble output
regression	$\hat{f}_b(x)$	$\frac{1}{B}\sum_{b=1}^{B}\hat{f}_b(x)$
binary classification	$\hat{p}_b(x)$ or class	average the probabilities then threshold, or majority vote
multiclass	class probability vector	argmax of the averaged class probabilities
ranking / policy	risk score, probability	combine with top-k, a threshold, a cost rule

6 Random Forests

6.1 Bagging alone can let every tree use the same strong variable

Correlated trees: if every tree starts its root split with the same strong variable, the prediction boundaries look alike even though the bootstrap samples differ. Problem: even with large B, a high ρ limits the variance reduction of the average.

6.2 Randomness over both observations and features

Algorithm: random forest

For $b = 1, \ldots, B$: draw a bootstrap sample; grow a tree in which, **at each node**, only m_{try} randomly chosen variables are allowed as split candidates; grow deep (minimum leaf size small). Aggregate as in bagging.

The bootstrap makes the trees' data different; the feature subsampling makes their first splits and structure different. Trees explore different explanatory paths, their correlation falls, and the averaging effect grows.

6.3 m_{try}

$m_{\text{try}} \in \{1, \ldots, p\}$; $m_{\text{try}} = p$ is bagging. Small m_{try}: more diverse trees, lower correlation, but individual trees may get weaker. Large: each tree stronger but trees more similar. Conventional starting values are $\sqrt{p}$ for classification and about $p/3$ for regression, but they depend on data and implementation; choose by OOB or CV and compute cost.

6.4 OOB scores are convenient but do not always replace CV

For observation i, average only the trees for which i was out of bag: $\hat{f}^{\text{OOB}}(x_i) = \frac{1}{|\mathcal{B}_i|}\sum_{b\in\mathcal{B}_i}\hat{f}_b(x_i)$, with $|\mathcal{B}_i| \approx 0.368B$; the OOB error is the average loss of these predictions.

Method	Strength	Caveat
OOB	fast internal estimate without a separate validation set	does not respect time/group structure; not independent validation
CV	compares preprocessing and candidates under the same rule	expensive; the pipeline must be repeated inside folds
final test	reports generalization of a fixed procedure	used once; not material for re-tuning

OOB is an excellent internal diagnostic for random forests, but does not remove the time/-group/leakage validation principles of Ch. 1.

7 Variable Importance Is Not Causal Importance

Impurity-decrease importance: sum over the forest the impurity reductions obtained by splits on each variable; fast and usually provided by default. **Permutation importance**: shuffle one column and see how much validation/OOB performance drops:

$$I_j = \text{Score}(D_{\text{eval}}) - \text{Score}(D_{\text{eval}}^{\pi(j)}),$$

where $D_{\text{eval}}^{\pi(j)}$ is the evaluation data with only column j permuted; for losses define the sign so that a "loss increase" is reported. Compute it on OOB or validation sets, not the final test, and record variability over repeated permutations.

High importance signals that the model's predictions *relied on* the variable. Correlated variables may share or substitute each other's importance: when one is shuffled, its substitute still carries the information, so its importance can look low. High-cardinality categorical variables are favored by impurity importance (more chances to split). Forbidden interpretation: high importance does not mean that changing the variable changes the outcome.

8 Comparison and Summary

Model	Randomness	Main strength	Main limitation
single tree	none or limited	rules are directly readable	high variance, unstable boundaries
bagging	bootstrap of rows	reduces variance of deep trees	gain limited if trees are correlated
random forest	rows + feature subsets	lower correlation, more stable prediction	hard to explain by one rule
boosting (next)	sequential residual / weight updates	reduces bias, strong prediction	more sensitive to tuning, overfitting, time structure

The key difference among tree ensembles is not "how many trees" but **how dissimilar the trees' errors are made**.

Summary

1. Trees split the input space by conditions and capture nonlinearity and interactions.
2. A deep single tree is expressive but has high variance and overfits.
3. Bagging averages bootstrap trees to lower variance; random forests add feature subsampling to lower correlation between trees.
4. OOB, importance and partial dependence are useful diagnostics, but do not replace independent validation or causal interpretation.
5. Trees must be compared with the linear baseline under the same split, CV and final-test rules.